

Lightweight Transformer Models for Real-Time Traffic Sign Recognition

A Comparative Study of Accuracy, Latency, and Model Efficiency

Abstract

This paper investigates whether compact Transformer-based vision models can provide competitive traffic sign recognition while meeting the latency and memory constraints of edge devices. We formulate the problem as multi-class image classification and compare a conventional convolutional baseline with a lightweight vision Transformer. The study proposes an evaluation framework covering classification accuracy, parameter count, inference latency, and robustness to illumination changes. A hypothetical experimental protocol is used to demonstrate how model efficiency should be assessed in practical intelligent transportation systems. The analysis suggests that architectural efficiency, input resolution, and augmentation policy jointly determine whether a model is suitable for deployment. The paper concludes with recommendations for selecting compact architectures when computational resources are limited.

Keywords: traffic sign recognition; vision Transformer; edge AI; computer vision; model compression

1. Introduction

Traffic sign recognition is a fundamental component of intelligent transportation systems and driver-assistance applications. A deployed recognition system must identify signs accurately while operating under changing illumination, weather, viewpoint, and image quality. Although deep convolutional networks have established strong performance, their computational requirements can become problematic on embedded hardware. Recent attention-based architectures offer a different mechanism for modeling visual relationships, but their benefits must be balanced against inference cost.

This paper studies the design trade-off between recognition quality and computational efficiency. Rather than optimizing accuracy alone, we treat latency and model size as first-class deployment constraints. The research question is: can a lightweight Transformer achieve useful recognition performance without imposing the computational burden associated with larger attention-based models?

2. Related Work

Earlier traffic sign recognition systems relied heavily on handcrafted color and shape features. Convolutional neural networks subsequently became the dominant approach because they learn hierarchical representations directly from images. More recent vision Transformers divide an image into patches and use self-attention to model relationships among these patches.

For edge deployment, lightweight CNN families, knowledge distillation, quantization, pruning, and efficient attention mechanisms are particularly relevant. These approaches demonstrate that high-capacity architectures do not necessarily provide the best engineering solution when memory, power, and latency are constrained.

3. Methodology

The proposed study uses a labeled traffic-sign image collection divided into training, validation, and test sets. Images are resized to a fixed resolution and augmented using random brightness changes, small rotations, horizontal-safe geometric transformations where semantically appropriate, and mild blur. Two models are evaluated: a compact CNN baseline and a lightweight vision Transformer.

Performance is measured using accuracy, macro-averaged F1 score, parameter count, and average single-image inference latency. Robustness is evaluated on subsets containing simulated low-light and motion-blur conditions. The models are trained under comparable optimization settings so that differences are primarily attributable to architecture.

4. Results and Discussion

A representative outcome of the proposed evaluation is that both models achieve strong recognition on clean images, while the lightweight Transformer provides improved performance on examples where relationships among multiple visual regions are important. However, the Transformer does not automatically dominate the CNN under every constraint. At very small input sizes, the CNN can remain faster and more memory-efficient.

The main implication is that model selection should be multi-objective. A model with one percentage point higher accuracy may be less useful if its latency doubles on the target processor. Quantization and knowledge distillation are therefore promising next steps for reducing the deployment gap.

5. Conclusion

This study presents a practical framework for evaluating lightweight Transformer models for traffic sign recognition. The analysis emphasizes accuracy, robustness, latency, and parameter efficiency rather than a single performance metric. The findings support the use of compact attention-based architectures when their additional computational cost is justified by improved robustness or accuracy. Future work should validate the comparison on physical edge devices and investigate post-training quantization and hardware-aware neural architecture search.

References

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